Yumeng He

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RESEARCH INTERESTS

My research interests lie in computer graphics and vision especially in physics-based animation, differentiable simulation, LLM-guided 3D scene synthesis and 3DGS.

EDUCATION

University of Southern California Aug 2024 – May 2026
Master of Science in Computer Science *Los Angeles, CA, USA*

- Cumulative GPA: 3.83/4.0
- Selected Coursework: Computer Graphics, Computer Animation and Simulation, 3D Graphics and Rendering

University of Toronto Sep 2019 – May 2024
Honours Bachelor of Science degree with High Distinction *Toronto, ON, Canada*

- Cumulative GPA: 3.71/4.0
- Computer Science and Mathematics double major
- Dean's List (top 10 %) — 2020 Summer, 2021 Winter, 2023 Summer, 2024 Winter
- Selected Coursework: Calculus I & II, Linear Algebra I & II, Ordinary Differential Equations, Abstract Mathematics, Complex Variables, Number Theory, Machine Learning, Artificial Intelligence

RESEARCH EXPERIENCE

AIVC Lab Visiting Student | full-time July 2025 – present
Supervisor: Prof. Chenfanfu Jiang *Los Angeles, CA, USA*

- Conducting research on interactive 3D scene-layout synthesis, investigating physics-aware placement and hierarchical generation strategies.
- Designing a novel LLM-guided pipeline that fuses large mesh databases with differentiable physics to automate scene assembly.
- Prototype under active development.

RELEVANT PROJECTS

Incompressible Fluid Simulation: A Comparison | C++, OpenGL Spring 2025

- Implemented and benchmarked five 2D incompressible fluid solvers—Stable Fluids, SPH, PIC, FLIP, APIC—on identical scenarios, reporting performance-vs-accuracy trade-offs.

Collision detection with penalty method and IPC | C++ Spring 2025

- Simulated a jello cube using a 3D mass-spring system.
- Modeled a network of Structural, Shear, and Bend springs to capture deformation behavior.
- Applied the penalty method for collision detection against boundaries and obstacles.
- Integrated line-search IPC; achieved 0 penetration ($> 1e-4$ relative gap) across 1k-step self-collision test.

Ray Tracing | C++ Fall 2024

- Developed a robust ray tracer in C++ capable of rendering both triangles and spheres using Phong shading, shadow rays, and recursive reflections for mirror-like surfaces.
- Implemented supersampling antialiasing, soft shadows from area lights, and the Moeller–Trumbore algorithm for efficient ray–triangle intersections, enhancing visual realism and performance.

Simulating a Roller Coaster | OpenGL Fall 2024

- Implemented first-person roller-coaster simulator using Catmull-Rom splines for track & Frenet frame for camera.
- Multi-pass OpenGL pipeline reaches 120 FPS at 1080p; added user speed control & auto-screenshot

Height Fields Using Shaders | OpenGL Fall 2024

- Built an interactive OpenGL pipeline with vertex and fragment shaders to generate real-time 3D terrain from heightmaps, supporting multiple rendering modes (points, lines, wireframe, smoothed surfaces).

- Integrated dynamic coloring, transformation/zoom controls, and automated screenshots/rotation features for an immersive, high-performance visualization experience.

Petpal | Full-Stack Fall 2023

- Led a four-person team to build a Python-based adoption platform with Django REST Framework, integrating MySQL for real-time data handling and advanced search filters.
- Implemented a responsive front end (HTML, CSS, JavaScript) and secure authentication to streamline the pet adoption process for seekers and shelters.

WORK EXPERIENCE

Software Developer Intern | full-time Aug 2022 – Aug 2023

HCL Canada Inc. *Toronto, ON, Canada*

- Utilized Jenkins to automate build and deployment processes and updated existing company code to address technical debt, significantly accelerating production releases and reducing human error.
- Containerized micro-services for x86_64 (xLinux) and IBM PowerPC (pLinux), enabling architecture-agnostic builds, ensuring consistent and scalable deployments.
- Integrated SonarQube into the development pipeline, documented a step-by-step usage guide, presented findings to cross-functional teams, and resolved 3 critical bugs plus 308 code smells (4 blockers, 41 major, 266 info).
- Developed a shell script to automatically clean up old logs every Sunday at 00:00 EST, which prevented clutter and improving system performance.

TECHNICAL SKILLS

Programming: C++, Python, JavaScript, HTML, CSS, MySQL
Graphics: OpenGL, GLSL, Eigen
ML: PyTorch
DevOps: Docker, Jenkins, Gradle, Ant, Shell
Languages: Mandarin (native), English (fluent)